

CompactObject: An open-source Python package for full-scope neutron star equation of state inference

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DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

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Submitted: 13 November 2024

Published: unpublished

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Summary

The equation of state (EOS) relates pressure, density, and composition in the interiors of neutron stars and determines observable quantities such as their masses, radii, and tidal deformabilities. **CompactObject** is an open-source Python framework for constructing or importing EOSs, calculating neutron-star structure, and constraining EOS parameters with Bayesian inference. Its modular design separates EOS generation, Tolman–Oppenheimer–Volkoff (TOV) integration, likelihood construction, sampling, and posterior visualization, so each layer can also be used independently. The package connects astrophysical measurements from radio timing, X-ray pulse-profile modelling, and gravitational waves with nuclear physics information from saturation properties, chiral effective field theory (χ EFT), and perturbative quantum chromodynamics (pQCD). It also reads cold EOS tables from CompOSE (CompOSE Core Team et al., 2022), allowing tabulated models to be analysed through the same interfaces as built-in models.

Statement of need

Inferring an EOS requires repeatedly translating model parameters into a thermodynamically consistent EOS, solving the stellar-structure equations, and evaluating heterogeneous observational and theoretical constraints. Comparing model families is particularly difficult if each is implemented with different unit conventions, crust matching, likelihoods, or sampling choices.

CompactObject supplies these operations in one reusable workflow. Its purpose is not to privilege one description of dense matter, but to make controlled comparisons between flexible parameterizations, nuclear empirical-parameter expansions, phenomenological field theories, and exotic compact-star models. The same likelihood and TOV machinery can therefore be

42 applied across model classes, while model-specific outputs such as particle fractions and nuclear
43 saturation properties remain accessible.

44 State of the field

45 Flexible piecewise-polytropic and speed-of-sound parameterizations are widely used to explore
46 broad EOS spaces. We refer to these as *parameterized EOS descriptions*, not metamodels.
47 Here, *nuclear metamodel* is reserved for an expansion of the energy per particle around nuclear
48 saturation density in terms of empirical isoscalar and isovector parameters (Margueron et al.,
49 2018), which is also implemented in CompactObject.

50 NEST provides an open-source nested-sampling framework with flexible core parameterizations
51 and low-density models informed by ab initio χ EFT calculations (Raaijmakers et al., 2025).
52 CompactObject is complementary: its particular emphasis is the use of a common inference
53 interface for those flexible descriptions alongside physics-motivated models whose couplings,
54 composition, and saturation properties can be inferred directly. This includes nonlinear and
55 density-dependent relativistic mean-field (RMF) models, strangeon matter, and quark matter.
56 Both approaches connect microscopic nuclear information and multimessenger data; they differ
57 mainly in the model libraries and interfaces they expose.

58 Software design and functionality

59 The EOSgenerators module implements piecewise-polytropic and speed-of-sound
60 parameterizations, a nuclear empirical-parameter expansion, nonlinear and density-dependent
61 RMF models (Malik et al., 2022; Tolos et al., 2017), a user-defined density-dependent coupling
62 interface, a strangeon EOS (Xu, 2003), and the MIT bag model (Chodos et al., 1974). It
63 also imports CompOSE and LAL-format tables. The TOVSolver module maps an EOS to
64 mass-radius and mass-radius-tidal deformability sequences and provides sound-speed and
65 maximum-central-density diagnostics.

66 InferenceWorkflow provides priors and likelihoods for mass, mass-radius, and tidal-
67 deformability measurements; nuclear saturation properties; pure-neutron-matter constraints
68 from χ EFT (Hebeler et al., 2013; Huth & others, 2022); and high-density pQCD consistency
69 (Gorda et al., 2023). Its nested-sampling workflow uses UltraNest (Buchner, 2021), while
70 documented notebooks show complete inference pipelines. Posterior comparison plots are
71 supplied by postprocessing. NumPy, SciPy, Numba, Jupyter, and corner.py provide the
72 principal numerical and interactive infrastructure (Foreman-Mackey, 2016; Kluyver et al., 2016;
73 Lam et al., 2015; Virtanen & others, 2020; Walt et al., 2011). Installation and notebook
74 smoke tests run in continuous integration for both pip and Conda environments.

75 Research impact

76 CompactObject has supported published Bayesian studies of nucleonic RMF models (Huang,
77 Raaijmakers, et al., 2024), hyperonic RMF models (Huang et al., 2025), strangeon matter
78 (Yuan et al., 2025), and first-order phase transitions (Huang & Sourav, 2025). A systematic
79 cross-comparison of covariant energy-density functionals has subsequently used the package
80 to apply identical astrophysical, χ EFT, and pQCD constraints across multiple model families
81 (Cartaxo et al., 2026). These applications demonstrate the intended use of the software for
82 reproducible inference and model comparison rather than only prospective functionality.

83 The source, examples, tests, and documentation are available from the project repository
84 (Huang et al., 2026), and versioned releases are archived on Zenodo (Huang, Malik, et al.,
85 2024). CompactObject is distributed under the GNU General Public License, version 3 or later
86 (GPL-3.0-or-later), consistently with the repository's LICENSE file.

AI usage disclosure

OpenAI Codex was used during the review revision to assist with language editing and bibliography auditing. The authors checked the resulting text against the source code and primary literature and take responsibility for the manuscript's content.

Acknowledgements

C.H., S.S., N.O., N.W., and J.G. acknowledge support from the Arts & Sciences Fellowship of Washington University in St. Louis. C.H. also acknowledges support from NASA grant 80NSSC24K1095. C.P. received support from Fundação para a Ciência e a Tecnologia (FCT), I.P., Portugal, under projects UIDB/04564/2020 (doi:10.54499/UIDB/04564/2020), UIDP/04564/2020 (doi:10.54499/UIDP/04564/2020), and 2022.06460.PTDC (doi:10.54499/2022.06460.PTDC). L.T. acknowledges support from CEX2020-001058-M (Unidad de Excelencia "María de Maeztu") and PID2022-139427NB-I00 financed by the Spanish MCIN/AEI/10.13039/501100011033/FEDER, EU; Generalitat de Catalunya contract 2021 SGR 171; Generalitat Valenciana contract CIPROM/2023/59; and CRC-TR 211, project 315477589 ("Strong-interaction matter under extreme conditions"). A.L.W. acknowledges support from ERC Consolidator Grant 865768 AEONS. A.C. acknowledges support from NSF grants DMS-2235457 and AST-2308111.

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